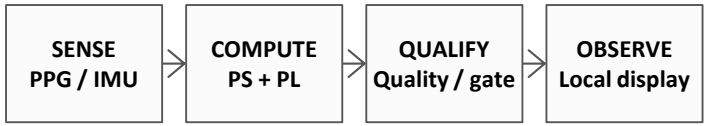
FPGA-Based Personal Health Trend Early Warning

— Trustworthy Heart Rate Observation for Multimodal Sensing

INTRODUCTION

Motion and contact changes disturb PPG. The prototype combines heart rate, waveforms and motion context to make signal quality observable.



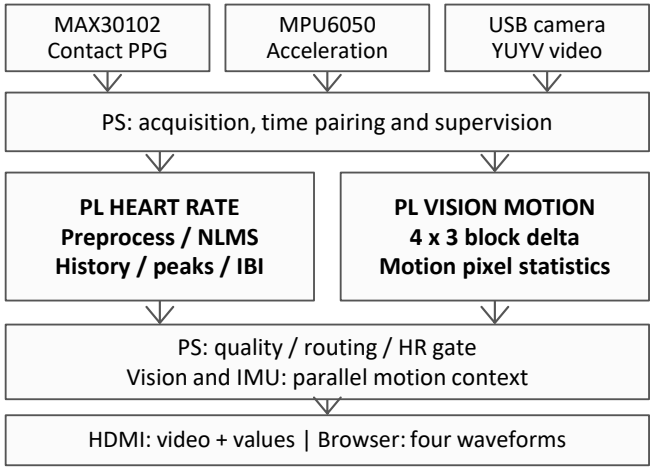
Observation Flow

Typical Use and Product Value

A local prototype for contact PPG observation. The browser shows four waveforms and quality; HDMI presents heart rate and motion context.

Key Engineering Contributions

1. Multimodal acquisition and time pairing.
2. PL preprocessing, NLMS, peaks and IBI.
3. Quality routing, qualified-IBI reuse and invalid-output withdrawal.
4. Custom supports integrate the board, display and camera; sensors use tape and hook-and-loop straps.
5. Eight reusable Skills support evidence-led engineering.



System Architecture

PL Heart Rate and Vision Processing

1. Preprocess, NLMS, peaks and IBI in PL.
2. Twelve blocks process image motion.
3. PS manages quality, routing and release.
4. Vision and IMU provide motion context.

Eight Skills and Human-Led LLM Work

People set scope; the model checks evidence.
Audit, sessions, timing, kernel verification;
Overlay, DMA, Notebooks and phase profiling.

CREATIVE DESIGN



PYNQ-Z2 board front



PYNQ-Z2 / AMD Zynq-7020

RESULT

Joint Run Test Results

Metric	Recorded result
Run window	Approx. 80.7 s
Processed / saved	2,021 / 2,021
Paired / unpaired	2,020 / 1
Unpaired record	Reason kept; HR withheld
Video	403 frames; target gaps 0
Stop	User stop; resources released

PPG/IMU: 25 Hz; video target: 5 fps

The joint run combines heart processing, vision observation and logging. Invalid records retain reasons and withhold values; pairing counts are separate from HR accuracy.

Joint PL main clock: 100 MHz; LUT 68.66%, BRAM 62.50%; WNS +0.016 ns, WHS +0.028 ns.

Same-input ARM component comparison: PS tail 5.04 → 4.23 ms/sample; Pipe path 9.24 → 8.05; messages 4 → 2 per sample. These are component gains, not whole-system speedup.